The documentation and process conversion measures necessary to comply with this revision shall be completed by 2 October 1999.

INCH-POUND

MIL-PRF-19500/443B
2 July 1999
SUPERSEDING
MIL-S-19500/443A
20 October 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPE 1N5719
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon pin diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500 (see 6.3.2).

1.2 Physical dimensions. See figure 1 (see 3.3).

1.3 Maximum ratings.

V_{RWM}	V_{RWM} non-repetitive	P_T 1/	T_{OP} and T_{STG}	$R_{\theta JC}$
V (pk)	V (pk)	mW	°C	°C/W
100	150	250	-65 to +150	500

1/ Derate 2.0 mA/°C above 25°C

1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

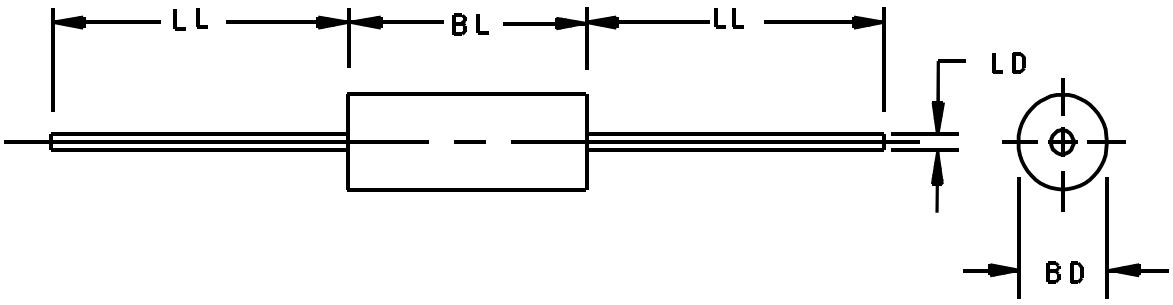
Limits	$V_{(BR)}$ I_R = 10 µA dc	R_S I_F = 100 mA dc 100 MHz	C_{VR} V_R = 100 V dc 100 kHz ≤ f ≤1 MHz	τ_{CL} I_F = 50 mA dc I_R = 250 mA dc
	V dc	Ω	pF	ns
Minimum	150			100
Maximum		1.25	.30	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
BD	.045	.076	1.14	1.93	3
BL	.100	.170	2.54	4.32	3
LD	.014	.016	0.36	0.41	4
LL	.975	1.5	24.77	38.10	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and BL. Heat slugs, if any , shall be included within this cylinder but shall not be subject to minimum limit BD.
4. Within this zone, lead diameter may vary to allow lead finishes and irregularities other than heat slugs.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1 Physical dimensions 1N5194UR, 1N5195UR, AND 1N5196UR.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

C_{VR} ..... Total capacitance with reverse bias.
τ_{CL} ..... Effective carrier lifetime.
P_{FM} ..... Peak forward power dissipation.
R_S ..... Residual series resistance.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, MIL-HDBK-6100, and herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.3.2 Polarity. The polarity shall be indicated by a contrasting color band to denote the cathode end. No color coding shall be permitted.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
9	Not applicable
11	I_{R1}, V_{F1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{R1} = ±100% of initial value or 250 nA dc, whichever is greater ΔV_{F1} = 100 mV (peak)

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750, minimum, I_O = 70 mA dc, V_R = 120 V(pk), f = 60 Hz.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1056	Test condition A.
B3	--1027	I_O = 70 mA dc minimum, V_R = 120 V(pk), f = 60 Hz. T_A = room ambient as defined in the general requirements of MIL-STD-750, minimum, T_J = 150°C minimum.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with the table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	--1027	I_O = 70 mA dc minimum, V_R = 120 V(pk), f = 60 Hz. T_A = room ambient as defined in the general requirements of MIL-STD-750, minimum, T_J = 150°C minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Effective carrier lifetime (τ_{CL}). Carrier lifetime may be measured using any of the following methods.

4.5.2.1 Reverse recovery time method. This test shall be conducted in accordance with MIL-STD-750, method 4031, condition B with the device installed in the test fixture, drawing C68001 or equivalent. The test parameters shall be: I_F = 50 mA, I_r = 30 mA, I(rec) = 3 mA.

4.5.2.2 Stored charge meter method. A pulse from a 20 kHz ± 200 Hz pulse generator and forward bias from a power supply shall be applied to the diode under test. Reverse current shall be 250 mA dc and forward bias shall be 50 mA dc. Read voltage on digital voltmeter and divide by 100 ohms to obtain average reverse current. Divide average reverse current by the product of pulse repetition frequency and bias current to obtain carrier lifetime (or equivalent) (see figure 2).

4.5.2.3 Stored charge meter method. Stored charge meter method (using commercial meter). Determine stored charge using meter manufacturer's instructions and a forward bias of 50 mA dc. Calculate the carrier lifetime by dividing the stored charge (coulombs) by the forward bias (amperes).

4.5.3 Series resistance (R_S). This test shall be performed utilizing an appropriate impedance analyzer such as the Hewlett Packard NO. 4191A or a vector network analyzer such as the Hewlett Packard No. 8753B (or equivalent).

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	I_F = 100 mA dc	V_{F1}		1.0	V dc
Reverse current	4016	DC method; V_R = 100 V dc	I_{R1}		250	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	150		V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Reverse current	4016	DC method; V_R = 100 V dc	I_{R2}		15	µA dc
Low temperature operation		T_A = -55°C				
Forward voltage	4011	I_F = 100 mA dc	V_{F2}		1.15	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	150		V dc
Subgroup 4						
Capacitance	4001	V_R = 100 V dc; 100 kHz ≤ f ≤1 MHz	C		0.3	pF
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Series resistance		I_F = 100 mA dc 100 kHz ≤ f ≤1 MHz	R_S		1.25	Ω
Effective carrier lifetime	----	I_F = 50 mA dc (see 4.5.2)	τ_{CL}	100		ns

1/ For sampling plan, see MIL-PRF-19500.

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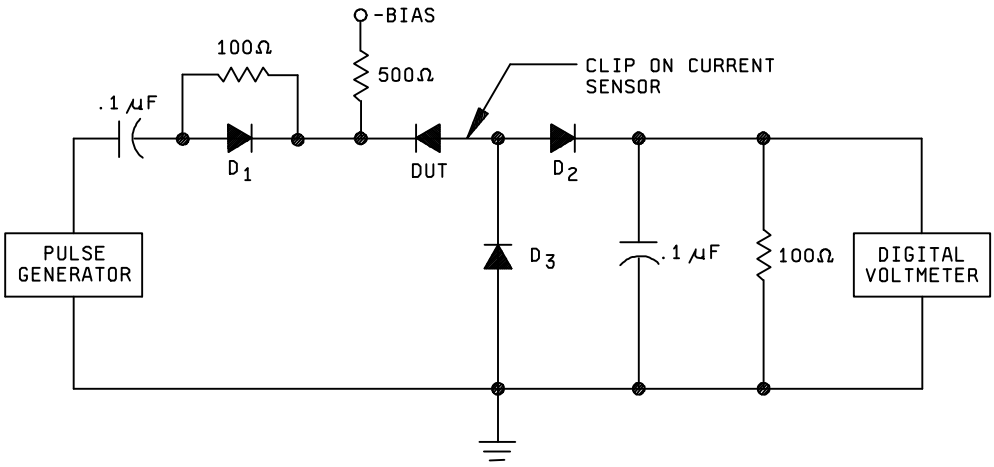
TABLE II. Groups B delta measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 100 mA dc	ΔV_{F1}	±100 mV dc maximum		
2.	Reverse current		DC method V_R = 100 V dc	$\Delta V_{(BR)1}$	±20 percent of initial value or ±50 nA dc (whichever is greater) change from initial group A reading.		

1/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 6, see table II herein, steps 1 and 2.

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NOTE:

D1 and 02 shall be 1N4456 high conductive diodes, or equivalent. For D3, use Hewlett Packard No. 5082-2835, or equivalent.

FIGURE 2. Stored charge meter method.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 19, 70, 80, 85, 99

Preparing activity:
DLA - CC

(Project 5961-2170)RP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/443B	2. DOCUMENT DATE 99/07/02

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING TYPE 1N5719 JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY				
a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC -LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvior, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

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